

Instructor: Date:

Learner: Regional Branch:

Program: Center:

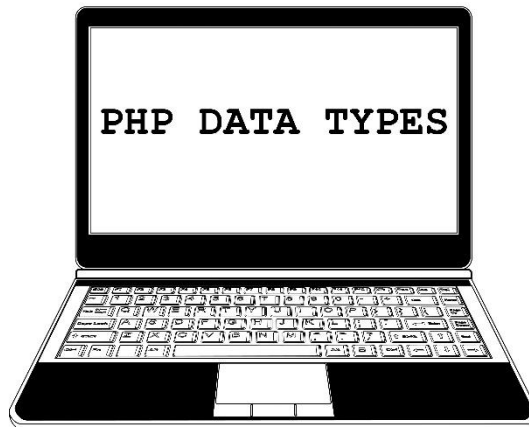
CONTENT

- Let's Start! • Let's Explore! • Let's Practice! • Extension Activity • Target Words • Learning Points



Let's Start!

Relevant Information About Data Types



Learning Outcome:

Understand information about web applications, their principles and basic concepts in a range of work contexts.
Describe applications for an operating system for personal computers in a range of work contexts.

Can Do Statements:

- I can list data types PHP supports.
- I can present information about data types like Boolean, array or object.
- I can talk about data types like string, integer or support.

Ask your partner:

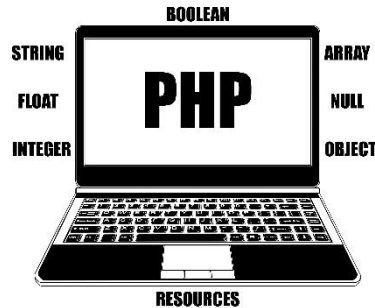
- What data types from other languages do you know? Why do we use them for?
- What do you think the advantages of PHP data types are?





Let's Explore!

Leo and Megan talk about the data types in PHP.



Instructions: Read the text below.

Megan: Wow, there are countless data types in PHP!

Leo: Megan, you are overreacting! There are only eight; String, Integer, Float, Boolean, Array, Object, NULL and... I always forget the last one...

Megan: Resource, right?

Leo: That one! Thanks! I've never used it because I understand it is an advanced type.

Megan: At first sight, no one would suppose PHP needs considerable expertise since beginners have been using it for a while now. Anyways, Leo, help me review them all!

Leo: Sure, let's start with string... what do you know about it?

Megan: String is a sequence of characters, like "this is the best", right?

Leo: That's basically it. What about integer?

Megan: Integer is a non-decimal number between -2,147,483,648 and its positive form. It has to have at least 1 digit, but it should never have a decimal point!

Leo: Wow, I wouldn't have thought you were so good at remembering numbers until now!

Megan: Hey, that's rude! For your information, I'm pretty good at numbers.

Leo: I'm sorry. Let's continue. What do you know about Boolean or Array?

Megan: Nothing!

Leo: Okay. Boolean represents two possible states, true or false, so it's very suitable for conditional testing. On the other hand, Array is useful to store multiple values in one variable.

Megan: Well, I thought they would be more difficult. What about NULL? NULL has appeared automatically many times when using PHP.

Leo: That's because NULL doesn't have a value, and if you create a variable with no value, it is automatically assigned as a NULL type.

Megan: Oh, that explains it. There is only one left; Object, what do we use that for?

Leo: It stores data and information on how to process that data. As a rule, it must always be explicitly declared, or it won't work.

Megan: Thanks, Leo. You have been of great help.

Leo: That's only because I have been working here for a while.





Let's Practice!

These people have been using different data types.

Instructions: Classify the sentences by writing the number of the sentence in the correct category.

- I've been using the data type which shows a sequence of characters.
- Annie has been practicing with the data type where non decimal numbers between - 2,147,483,648 and 2,147,483,648 are used.
- We have been following the rules for this data type where at least one digit is needed and no decimal point can be written.
- Hans has used single and double quotes for this data type.
- They have written the following line: \$ x = 345, for this data type.

String	Integer



Let's Practice!

Sophie describes some data types.

Instructions: Select the correct answer.

- I would use _____ if I were doing conditional testing.
 - Array
 - Boolean
 - Float
 - Integer
- They would use _____ if they needed to store multiple values.
 - String
 - Object
 - Array
 - Integer
- I would use _____, but I don't need to store information on how to processes that data.
 - Boolean
 - Integer
 - String
 - Object
- You would have written the object data type correctly if you had first declared a _____ of object.
 - keyword
 - class
 - property
 - method





Extension Activity

Learning More About Data Types.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more information about data types.
- Choose 4 types, and make a chart in which you include all their functions and characteristics.
- Share your chart with the rest of the class.

Self-Assessment

Yes

No

- I look for more information about data types.
- I make a chart about 4 data types including functions and characteristics.
- I share my chart with the rest of the class.

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Target Words

- **countless** (adjective): referring to impossible to count; a lot of something
e.g. Wow, there are countless data types in PHP!
- **at first sight** (expression): apparently, a first impression
e.g. At first sight, no one would suppose PHP needs considerable expertise since beginners have been using it for a while now.
- **overact** (verb): to exaggerate an action
e.g. Megan, you are overreacting! There are only eight.
- **appear** (verb): to exist; to act.
e.g. NULL has appeared automatically many times when using PHP.
- **explicitly** (adverb): referring to directly; straight forward
e.g. As a rule, it must always be explicitly declared, or it won't work.
- **suppose** (verb): to assume; to expect
e.g. No one would suppose PHP needs considerable expertise since beginners have been using it for a while now.



Learning Points

Modal Verbs for Speculating: Would

We use **would** to talk about hypothetical events.

When we want to talk about events in the future that are hypothetical, we can use the modal verb **would**. These events are not very likely to take place; that's why, it's necessary to express probability and not certainty.

Let's take a look at the structure:

Subject + **would** + verb in base form + complement

Examples:

*She **would** work as an engineer, but she doesn't know the PHP data types.*

*This program **would** run better with PHP object, but there is an object declared.*

Examples

- *I wouldn't have thought you were so good at remembering numbers until now!*
- *I thought they would be more difficult.*

Null Value and Resource Data Type in PHP

There are two special data types in PHP: **Null value** and **Resource data**.

NULL Value

Null is special because it can only have one value. As its name suggests that value is NULL. You could think it is the same as having a value of zero, but it's not.

When you create a variable, but do not assign a value to it; automatically, it is assigned a **Null** value, which is very different from writing zero as a value.

Also, when you set another variable to be **NULL**, the variable data will be emptied.

PHP Resource

It is not exactly a data type. PHP Resource stores a reference to operations and resources that are external to PHP.

A good example of this data type is a database call.

